
Essex Summer School 2026: Introduction to QTA

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 - Office hours: by appointment (via Zoom)
 - Meetings: Daily 10:00–13:30 (BST)
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Course introduction

With the massive growth of digital (and digitized) text data, quantitative text analysis (QTA) has become an increasingly important tool in the social sciences. This course offers a hands-on introduction to key methods and tools used in QTA for studying a wide range of social and political phenomena. Participants will learn about the theoretical foundations of QTA, including different approaches to representing and modeling text, as well as practical applications in empirical research. We will also spend time on new developments in QTA, with a specific eye towards using large language models (LLMs) for text annotation.

The course combines lectures with interactive coding sessions in R to guide participants through the entire research pipeline, from data collection to interpretation.

Learning Outcomes

Participants will understand fundamental concepts in quantitative text analysis research design such as inter-coder agreement, reliability, validation, accuracy, and precision. Participants will learn to convert texts into informative feature matrices and to analyze those matrices using statistical methods. Participants will learn to apply these methods to a text corpus in support of a substantive research question. Furthermore, participants will be able to critically evaluate (social science) research that relies on quantitative text analysis methods.

Participation and communication

I expect that you will have prepared for our meetings, having read required papers, and being ready to discuss your questions, criticisms and thoughts. To facilitate communication and interaction we will make use of a dedicated Slack channel at <https://essqta26.slack.com> for which I will send you an invitation via email.

Literature

Throughout the course we will read papers from political science and other social sciences. For further background on quantitative text analysis and natural language processing I recommend the following books:

- Daniel Jurafsky and James H. Martin (2026). *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*. 3rd edition. <https://web.stanford.edu/~jurafsky/slp3/>

- Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze (2008). An Introduction to Information Retrieval. New York: Cambridge University Press. <https://nlp.stanford.edu/IR-book/information-retrieval-book.html>
- Grimmer, J., Roberts, M.E. and Stewart, B.M. (2022). Text as data: A new framework for machine learning and the social sciences. Princeton University Press.
- van Atteveldt, W., Trilling, D. and Calderon, C.A., (2022). Computational Analysis of Communication. John Wiley & Sons. <https://v2.cssbook.net/>

Software

In this course we will use R. Students will need to have R and Positron installed on their computers / laptops. Students who have not used R at all are advised to work their way through the beginner courses listed on <https://education.rstudio.com/>. Another great resource is R for Data Science by Hadley Wickham and Garrett Grolemund. This book is available at <https://r4ds.had.co.nz/>.

Course outline

** This outline serves a general plan for the course; deviations (announced) may be necessary. To keep the workload manageable we'll stick to 2 readings (sometimes 3) a day but during our meetings we'll discuss other papers as well.*

Day 1 – 6 July

- What is quantitative text analysis? And what is quantitative text analysis in the age of LLMs? What will you learn in this course?
 - **Required reading:**
 - * Benoit, K. (2020). Text as Data: An Overview. Handbook of Research Methods in Political Science and International Relations. Ed. by L. Curini and R. Franzese. Thousand Oaks: Sage: 461–497.
 - * Grimmer, J., Roberts, M.E., and Stewart, B.M. (2022). Text as Data: A New Framework for Machine Learning and the Social Sciences. Princeton: Princeton University Press: chapter 4.

Day 2 – 7 July

- Core assumptions in quantitative text analysis. Representations of text. Preprocessing, feature selection and input design.
 - **Required reading:**
 - * Baden, C., Pipal, C., Schoonvelde, M. & van der Velden, M.A.C.G., (2022). Three Gaps in Computational Text Analysis Methods for Social Sciences: A Research Agenda. *Communication Methods and Measures*, 16(1): pp. 1–18.
 - * Benoit, K., Watanabe, K., Wang, H, Nulty, P., Obeng, A., Müller, & Matsuo, A. (2018). Quanteda: An R package for the quantitative analysis of textual data. *Journal of Open Source Software*, 3(30), 774.

Day 3 – 8 July

- Advanced text representations: embeddings, contextual embeddings, and LLM representations. Comparing representations.
 - **Required reading:**
 - * Grimmer, J., Roberts, M.E., and Stewart, B.M. (2022). Text as Data: A New Framework for Machine Learning and the Social Sciences. Princeton: Princeton University Press: chapter 8.
 - * Rodriguez, P.L. and Spirling, A., (2022). Word embeddings: What works, what doesn't, and how to tell the difference for applied research. *The Journal of Politics*, 84(1): pp.101–115.

Day 4 – 9 July

- Dictionary construction and validation. How are dictionaries still relevant in the age of LLMs?
 - **Required reading:**
 - * Grimmer, J., Roberts, M.E., and Stewart, B.M. (2022). Text as Data: A New Framework for Machine Learning and the Social Sciences. Princeton: Princeton University Press: chapter 16.
 - * Rauh, C., 2018. Validating a sentiment dictionary for German political language—a workbench note. *Journal of Information Technology & Politics*, 15(4): pp.319–343.

Day 5 – 10 July

- Supervised document classification through human and LLM-assisted coding. Evaluating classifiers and annotations.
 - **Required reading:**
 - * Daniel Jurafsky and James H. Martin (2026). Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition. 3rd edition: Chapter 4
 - * Gilardi, F., Alizadeh, M., & Kubli, M. (2023). “ChatGPT Outperforms Crowd-Workers for Text-Annotation Tasks”. *Proceedings of the National Academy of Sciences of the United States of America* 120 (3): e2305016120.

Day 6 – 13 July

- Supervised, semi-supervised, unsupervised, and instruction-based approaches to placing text on an underlying dimension. Discussing their pros and cons.
 - **Required reading:**
 - * Schwemmer, C. and Wieczorek, O., (2020). The methodological divide of sociology: Evidence from two decades of journal publications. *Sociology*, 54(1), pp.3–21.
 - * Watanabe, K., (2021). Latent semantic scaling: A semisupervised text analysis technique for new domains and languages. *Communication Methods and Measures*, 15(2), pp.81–102.

Day 7 – 14 July

- Understanding topic models. Discussing their pros and cons.
 - **Required reading:**
 - * Roberts, M et al. (2014). Structural topic models for open-ended survey responses. *American Journal of Political Science*, 58(4), pp. 1064–1082.
 - * Grimmer, J., Roberts, M.E., and Stewart, B.M. (2022). Text as Data: A New Framework for Machine Learning and the Social Sciences. Princeton: Princeton University Press: chapter 13.

Day 8 – 15 July

- New forms of textual data: multilingual text, speech-to-text, multimodal data.
 - **Required reading:**
 - * Proksch, S.O., Wratil, C. and Wäckerle, J., (2019). Testing the validity of automatic speech recognition for political text analysis. *Political Analysis*, 1–21
 - * Licht, H. and Lind, F., (2023). Going cross-lingual: A guide to multilingual text analysis. *Computational Communication Research*, 5(2) pp. 1–31.
 - * Birkenmaier, L., Lechner, C.M. and Wagner, C. (2024). The search for solid ground in text as data: A systematic review of validation practices and practical recommendations for validation. *Communication Methods and Measures*, 18(3): pp. 249–277.

Day 9 – 16 July

- New developments in supervised machine learning. Weak supervision. Transfer learning.
 - **Required reading:**
 - * Laurer, M., Van Atteveldt, W., Casas, A. & Welbers, K., (2024). Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI. *Political Analysis*, 32(1), pp. 84–100.
 - * Kroon, A., Welbers, K., Trilling, D., & van Atteveldt, W. (2024). Advancing automated content analysis for a new era of media effects research: The key role of transfer learning. *Communication Methods and Measures*, 18(2), pp. 142–162.
 - * S. Wankmüller (2024). Introduction to Neural Transfer Learning with Transformers for Social Science Text Analysis. *Sociological Methods & Research* 53 (4): 1676–1752.

Day 10 – 17 July

- LLMs in QTA: annotation, classification, extraction, validation, and reporting. Concluding remarks
 - **Required reading:**
 - * Bail, C.A., (2024). Can Generative AI improve social science? *Proceedings of the National Academy of Sciences*, 121(21) p.e2314021121.
 - * Benoit, K., De Marchi, S., Laver, C., Laver, M. and Ma, J. (2025). Using large language models to analyze political texts through natural language understanding. *American Journal of Political Science*: pp 1-17.
 - * Laurer, M., van Atteveldt, W., Casas, A., & Welbers, K. (2025). On measurement validity and language models: Increasing validity and decreasing bias with instructions. *Communication Methods and Measures* 19(1), pp 46–62.